

Software Project Management Plan

AI Task Manager

10/5/25

Bit Crew

Computer Science Dept/ CSUN

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Revisions

Version	Primary Author	Description of Version	Date completed
1.0	All team members	Initial version	10/5/25
1.1	Ashton Rosales	Final Version	4/12/26
1.2			
1.3			
1.4			
1.5			

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1. Introduction

1.1 Project Overview

Purpose

The purpose of this project is to develop a mobile app using Expo + React Native, gaining hands-on experience in UI design, data management, and smartphone integration. Throughout the development process, we will practice breaking down complex tasks, adapting to challenges, and managing changing requirements. Additionally, we will strengthen our problem-solving skills by addressing bugs, design issues, and technical limitations—valuable experience for future careers in software development.

Scope

We are planning on implementing the following features so far:

- Reminders for due dates
- A calendar interface or integration with built in calendar app on the phone
- Task manager
- AI assistant (guides users through task and schedule creation)

Assumptions and constraints

- Assumption is that the user has access to either an android or ios device (smartphone or tablet) with app support.
- Limitations/constraints are:
 - Limited prior experience with React Native or mobile development
 - Limited knowledge of data storage
 - Possible compatibility issues between iOS and Android
 - App must remain simple enough to finish within given time frame
 - Limited user testing as testers are likely just going to be the team members

1.2 Literature Review

- Existing productivity tools such as Akiflow demonstrate that combining a calendar interface with AI-assisted scheduling significantly improves how users manage their time. Akiflow's time-blocking approach, where tasks are assigned to specific time slots, directly influenced our calendar and task placement design.^{1 2 3}
- Expo + React Native was selected as the frontend framework because it allows a single codebase to be deployed on both Android and iOS while providing access to native device APIs. This reduces development overhead compared to maintaining separate native codebases.^{4 5}
- FastAPI was chosen for the backend due to its high performance, built-in data validation via Pydantic, and automatic API documentation generation. Its Python foundation also aligns with the team's existing backend knowledge and integrates naturally with AI tooling.⁶
- SQLAlchemy with SQLite provides a lightweight, file-based database layer that requires no separate database server to run. This suits the project's scope and simplifies local

¹ "Time-Blocking Digital Planner & Calendar," Akiflow. Available: <https://akiflow.com/>

² "Mobile App," Akiflow. Available: <https://how-to-use-guide.akiflow.com/mobile-app>

³ "Time Blocking," Toggl Blog. Available: <https://toggl.com/blog/time-blocking>

⁴ "React Native," reactnative.dev. Available: <https://reactnative.dev>

⁵ "Expo," expo.dev. Available: <https://expo.dev>

⁶ "FastAPI," tiangolo. Available: <https://fastapi.tiangolo.com>

development and deployment.⁷

- Ollama enables large language models to run locally on the developer's machine without requiring cloud API access. This reduces latency, eliminates API costs, and keeps user data local. The qwen2.5:1.5b model was selected for its small footprint while remaining capable enough for conversational, task-driven guidance.⁸

2. Project Organization

2.1 Roles and Responsibilities

Team Member	Roles	Email
Ashton Rosales Leo Aghazarian	Backend developer(Databases)	ashton.rosales.348@my.csun.edu leo.aghazaryan.483@my.csun.edu
Shree Vidhya Nuthati Delilah Lazaritt	Frontend developer	shree.nuthati.668@my.csun.edu delilah.lazaritt.990@my.csun.edu
Levi De Luis Sanchez Brian Nguyen	Backend developer	levi.de-luis.sanchez.411@my.csun.edu brian.nguyen.079@my.csun.edu

2.2 Tools and Techniques

- **JIRA** - Project management, sprint planning, and issue tracking
- **GitHub** - Version control and code collaboration
- **PyCharm** - IDE for Python and AI development
- **React Native** - Frontend framework for building user interface
- **WebStorm** - IDE for frontend and React Native development
- **Discord** - Team communication and coordination
- **Akiflow (Reference)** - Reference for AI-based scheduling and productivity integration (similar to Aki AI)
- **Figma** - UI/UX design and prototyping tool
- **FastAPI + Uvicorn** - Backend framework, server
- **Ollama** - Local LLM runtime, qwen2.5:1.5b
- **Zustand** - Frontend state management
- **Expo** - Mobile development platform for React Native; simplifies building, testing, and deploying on Android and iOS
- **python-jose + passlib/bcrypt** - For secure password hashing and JWT token generation/verification

⁷ "SQLAlchemy Documentation," SQLAlchemy. Available: <https://docs.sqlalchemy.org>

⁸ "Ollama," Ollama. Available: <https://ollama.com>

- **expo-notifications** - Expo library used to schedule and dispatch local push notifications on Android devices.
- **SQLAlchemy** - Core ORM, data access layer for the backend

3. Project Management Plan

3.1 Tasks

Phase/Task#	Task Description	Due Date
0	Research – GitHub, Setting Up IDEs	10/12/25
1-1	Requirements Gathering	10/19/25
1-2	Frontend: Learn React Native with Expo, Backend: Learn Python/FastAPI Database: Learn SQL Alchemy 2.0	10/26/25
2-1	Design - UML Class Diagrams and User Flow Diagrams	11/2/25
3-1	Construction – Developing the code	11/3/25
3-2	Create UI	11/30/25
3-3	Create Calendar	11/30/25
3-4	Implement AI Assistant (Ollama Integration)	11/30/25
3-5	Authentication	11/30/25
3-6	Complete Construction Phase	11/30/25
4-1	Testing – Unit Testing	12/7/25
5-1	Deployment – Implement our code	12/14/25
6-1	Feedback – Peer Review	12/14/25

3.2 Assignments

Week#	Deliverables/Progress
3	Progress Report #1
4	Progress Report #2
4	Project Proposal Presentation
6	Progress Report #3
6	Software Project Management Plan (SPMP)

8	Progress Report #4
10	Progress Report #5
11	Software Requirement Specification (SRS)
12	Progress Report #6
12	Code Review #1 Artifacts
14	Software Design Document (SDD)
14	Progress Report #7
16	Project Presentation
16	Project Source Code
16	Progress Report #8

3.3 Timetable

- Research
- Phase 1-Requirements Gathering
- Phase 2- Design
- Phase 3- Construction
- Phase 4- Testing
- Phase 5- Deployment & Feedback

Additional Material

Definitions, Acronyms and Abbreviations

- AI - Artificial Intelligence
- IDE - Integrated Development Environment
- UI - User Interface
- LLM - Large Language Model
- JWT - JSON (JavaScript Object Notation)
Web Token